ROHIT VEMPATI

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RESEARCH INTERESTS

My research interests lie in understanding how data-driven systems behave under uncertainty, particularly in settings where temporal dynamics and interacting components influence outcomes. I am interested in studying not only predictive performance, but also how system behavior changes under input perturbations, shifting conditions, and sequential dependencies.

- **System Behavior in Data-Driven Models**: Analyzing how model outputs change under small variations in input, with a focus on sensitivity, stability, and consistency across different conditions
- **Time Series and Sequential Systems**: Modeling systems where decisions evolve over time, with emphasis on how early-stage uncertainty propagates through downstream outcomes
- **Robustness and Sensitivity Analysis**: Studying how preprocessing, feature construction, and environmental factors influence model behavior beyond aggregate performance metrics
- **Structured Reasoning in ML Systems**: Exploring ways to move from empirical evaluation toward more structured reasoning about system behavior, particularly in environments with uncertainty and evolving inputs

EDUCATION

Wayne State University — **Detroit, MI**
M.S. Data Science & Business Analytics (Advanced Analytics) - GPA:3.81 — *August 2023 – December 2024*
***Graduate coursework**: Neural Networks; Intelligent Analytics; Data Mining: Algorithms & Applications; Computational Platforms for Data Science; Statistics for Data Science; Data Science Strategy & Leadership.*

NIIT University
B.Tech. Computer Science Engineering (concentration in Data Science) — *July 2017 – July 2021*

SKILLS

- **Programming & Data Science:** Python (NumPy, Pandas, scikit-learn, LightGBM, OpenCV), SQL (SAP, Spark, BigQuery), R.
- **Machine Learning & AI:** Regression, classification, clustering, ensemble methods, time-series forecasting, neural networks, computer vision.
- **Big Data & Cloud:** Azure Databricks, Azure Data Lake, Spark SQL, GCP BigQuery, Snowflake.
- **BI & Visualization:** Power BI (DAX, Power Query, RLS, parameters), Tableau, Tableau Prep, Advanced Excel.
- **Statistics & Research Methods:** Hypothesis testing, probability distributions, regression diagnostics, scenario analysis, sensitivity analysis.
- **Enterprise Tools & Practices:** SAP, Alteryx, Informatica IICS, Collibra, Git/GitLab, Agile.
- **Optimization & OR:** IBM CPLEX/OPL, linear & integer programming, dynamic programming, heuristics.

RESEARCH EXPERIENCE

Graduate Research Assistant – Intelligent Analytics Lab, Wayne State University — *September 2024 – Present*
*Advisor: **Prof. Ratna Babu***

- **Project Objective:** To design and implement a LiDAR-based robotic scanning pipeline for automated disassembly and remanufacturing, with the aim of constructing data-driven representations of physical components that support disassembly planning and simulation-based analysis, while enabling evaluation of **model behavior under varying environmental conditions**.
- **Methodology:** Developed data pipelines to process and analyze **50+ LiDAR scans**, leveraging Python, NumPy, and Pandas for feature engineering of spatial and geometric properties. Applied computer vision (OpenCV) to evaluate **1,200+ detection outputs** under diverse lighting and background conditions. Designed benchmarking protocols, rolling evaluations, and visualization frameworks to measure **robustness and sensitivity to environmental variability**, with emphasis on how **preprocessing and feature construction influence downstream model behavior**.

- **Findings:** Improved detection consistency by **15–20%**, demonstrating that LiDAR-CV integration can mitigate environmental variability. Observed that performance degradation was often driven by **sensitivity to input perturbations and representation choices**, rather than model architecture alone.
- **Specific Contributions:** Led the end-to-end research cycle from raw scan processing and feature engineering to system-level evaluation. Developed methodologies to **identify failure modes under varying conditions** and assess **model reliability beyond static performance metrics**. Authored technical documentation emphasizing **reproducibility and systematic evaluation of modeling assumptions**, and produced visualization dashboards (Matplotlib, Seaborn) to analyze system behavior and guide iterative improvements.

RESEARCH & PROFESSIONAL EXPERIENCE

Tenneco **Southfield, MI**
Data Scientist *May 2024 – September 2024*

- **Time Series Modeling & Forecast Behavior Analysis:** Developed **time series forecasting models for 16,000+ SKUs**, analyzing **temporal patterns, seasonality, and demand variability** across heterogeneous product lines. Evaluated how different approaches respond to short-term fluctuations versus structural changes, identifying **trade-offs between responsiveness and stability over time**. Improved predictive accuracy by **~35%**, while recognizing that similar aggregate metrics can mask differences in model behavior across time horizons.
- **Large-Scale Temporal Data Systems & Pipeline Design:** Built scalable Python-based pipelines on Azure Databricks to process **720K+ weekly time-dependent records**, enabling consistent integration of multi-source data for large-scale model evaluation. Designed transformation and validation workflows to ensure **data consistency, temporal alignment, and robustness**, supporting reliable analysis of evolving demand patterns.
- **Comparative Modeling & Assumption-Driven Analysis:** Evaluated **13 forecasting techniques**, benchmarking predictive performance, scalability, and behavior across demand scenarios. Focused on how **assumptions related to trend, seasonality, and noise structure** influence forecast outputs and downstream planning decisions, contributing to the selection of a hybrid modeling approach.

Quantiphi Inc.
Analyst – AI/ML *July 2021 – July 2023*

- **Demand Modeling & Temporal Analysis:** Led forecasting and modeling efforts for EdTech/FinTech datasets, analyzing **demand and adoption patterns over time** using behavioral and external signals to understand how engagement evolved across user segments.
- **Data Integration & Analytical Systems:** Built analytical workflows integrating third-party and internal CRM datasets, enabling analysis of **temporal trends and regional variability**, and supporting data-driven decision systems that contributed to **~30% improvement in client acquisition outcomes**.
- **Model Development & Feature Validation:** Collaborated with data science and engineering teams to prototype ML features and validation pipelines, focusing on **data consistency, feature reliability, and interpretability of model outputs** in early-stage deployments.
- **Applied AI & Research-Oriented Development:** Contributed to the development of AI-driven systems (AI tutor tool), investigating **adaptive feedback mechanisms, personalization strategies, and model-driven content delivery**, with emphasis on scalable learning systems.

WaysAhead Global
Data Science Intern *January 2021 – July 2021*

- **Customer Segmentation & Behavioral Modeling:** Designed and implemented an end-to-end **RFM segmentation pipeline**, focusing on identifying customer behavior patterns across purchase cycles and engagement stages.
- **Feature Engineering & Segment Stability:** Engineered recency, frequency, and monetary features, and evaluated **segment consistency across time windows**, analyzing how customer grouping changed with evolving behavioral patterns.
- **Cohort Analysis & Retention Patterns:** Conducted **cohort and retention analysis**, identifying trends in repeat behavior and engagement across customer groups over time.
- **Sensitivity Analysis & Pattern Interpretation:** Analyzed how **segmentation thresholds and input assumptions influenced segment definitions**, and developed dashboards to visualize **segment evolution and temporal behavioral patterns**.

References available on request